Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA101-1 **Max Marks: 100**

Course Name: FOUNDATIONAL MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Reformulate statements from common language to formal logic using the rules of propositional and predicate calculus and assess the validity of arguments.

CO2: formulate and interpret statements presented and determine their validity by applying the rules and methods of propositional logic.

CO3: construct elementary proofs using ordinary and strong induction in the context of studying the properties of recursion.

CO4: apply basic counting principles including the pigeonhole principle and rules for counting permutations and combinations.

CO5: apply the logical structure of proofs and work symbolically with connectives and quantifiers to produce logically valid, correct and clear arguments

Section A **3*8=24**

1)	Show that among any group of five (not necessarily consecutive) integers, there are two with the same remainder when divided by 4.	**3**	(CO4, L4)
2) a)	State and prove the Modus Ponens law without using truth table.	**3**	(CO1, L1)
	OR		
b)	Show that " I must have watched TV in the evenings " is a valid conclusion from the premises "If I study, I will not fail in the examination", "If I do not watch TV in the evenings, I will study" and "I failed in the examination".	**3**	(CO2, L1)
3)	How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits? Explain the mathematical principle used to get the answer.	**3**	(CO4, L3)
4)	Construct a truth table for compound proposition .	**3**	(CO1, L1)
5) a)	Show that if and are real numbers and , then there is a unique real number such that . ¬(p →q) →¬q (p →q) (¬p →r) P(x) x Q(x) x x(P(x) →Q(x)) x(P(x) →Q(x)) x(P(x) Q(x)) n 5n + 6 a b a =0 r ar + b = 0	**3**	(CO3, L3)
	OR		
b)	If is a positive integer, then construct a formula for and prove it by mathematical induction.	**3**	(CO3, L3)
6)	Define POSET of a set with an example.	**3**	(CO2, L3)

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7) a)	Show that the set of odd positive integers is a countable set.	**3**	(CO2, L2)
	OR		
b)	How can a directed graph of a relation on a finite set be used to determine whether the relation is reflexive, symmetric?	**3**	(CO3, L2)
8)	How can a directed graph of a relation on a finite set be used to determine whether the relation is reflexive, symmetric?	**3**	(CO3, L2)
	Section B		**8*7=56**
9)	Prove by mathematical induction that, for all positive integers ,	**8**	(CO3, L2)
10) a)	How many positive integers between 5 and 31: (i) a) are divisible by 3? Which integers are these? (ii) are divisible by 4? Which integers are these? (iii) are divisible by 3 and by 4? Which integers are these?	**8**	(CO4, L3)
	OR		
b)	In a survey of 60 people it was found that, 25 read weekly magazines, 26 read fortnightly magazines, 26 read monthly magazines, 9 read both weekly and monthly magazines,11 read both weekly and fortnightly magazines, 8 read both monthly and fortnightly magazines and 3 read all 3 magazines. Develop the Venn diagram and hence determine the number of people who read a) at least one of the three magazines. b) exactly one magazine.	**8**	(CO1, L5)
11)	Determine the following congruences are solvable or not. (i) 8 ≡ 10 (6) (ii) 2 ≡ 3 (4) (iii) 4 ≡ 7 (5) (iv) 2 ≡ 3 (5)	**8**	(CO4, L4)
12) a)	Prove that if is an integer, these four statements are equivalent: (i) is even (ii) is odd (iii) is odd (iv) is even.	**8**	(CO3, L3)
	OR		
b)	1 If is a real number, prove that 2 = + + . 2	**8**	(CO2, L1)
13)	Prove by at least three different methods: “If is even, then is even”.	**8**	(CO3, L3)
14) a)	1 If is a real number, prove that 2 = + + . 2	**8**	(CO2, L1)
	OR		
b)	Determine the following congruences are solvable or not. (i) 8 ≡ 10 (6) (ii) 2 ≡ 3 (4) (iii) 4 ≡ 7 (5) (iv) 2 ≡ 3 (5)	**8**	(CO4, L4)
15)	Prove that if is an integer, these four statements are equivalent: (i) is even (ii) is odd (iii) is odd (iv) is even.	**8**	(CO3, L3)
	Section C		**10*2=20**
16)	Draw the Hasse diagram for the poset (, |), where the set S consist {2, 4, 6, 9, 12, 18, 27, 36, 48, 60, 72}. Hence answer the following a) Find the maximal and minimal elements. b) Is there a greatest and least element? if so write it. c) Find all upper bounds of {2, 9} and write the least upper bound of {2, 9}, if it exists. d) Find all lower bounds of {60, 72} and write the greatest lower bound of {60, 72}, if it exists	**10**	(CO1, L3)
17) a)	State the Chinese Remainder theorem, and hence solve the following system of linear congruences, ≡ 2 (3), ≡ 3 (5), ≡ 2 (7).	**10**	(CO4, L3)
	OR		

b)	Draw the Hasse diagram for the poset (, |), where the set S consist {2, 4, 6, 9, 12, 18, 27, 36, 48, 60, 72}. Hence answer the following a) Find the maximal and minimal elements. b) Is there a greatest and least element? if so write it. c) Find all upper bounds of {2, 9} and write the least upper bound of {2, 9}, if it exists. d) Find all lower bounds of {60, 72} and write the greatest lower bound of {60, 72}, if it exists	**10**	(CO1, L3)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create